CSCI – 310 Prep. Systems Software / 593 Intr. Operating Systems

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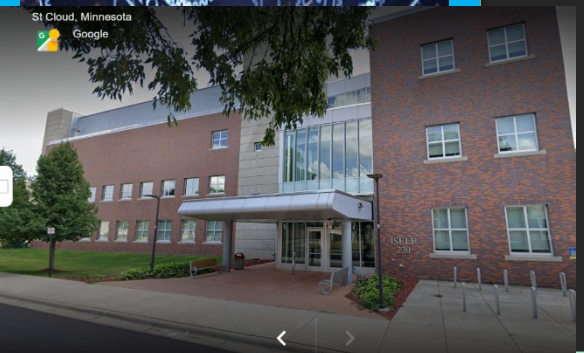

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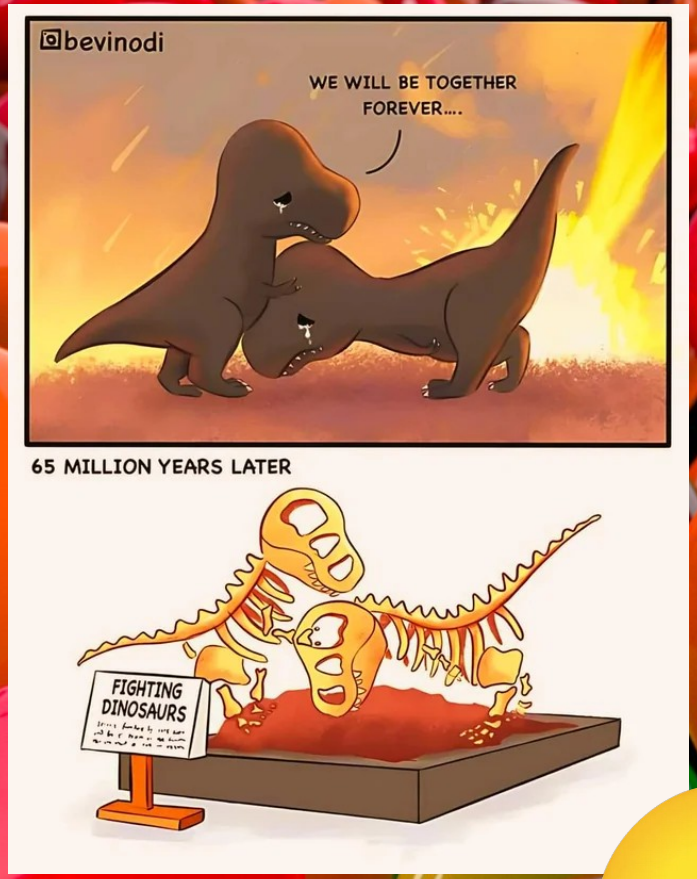
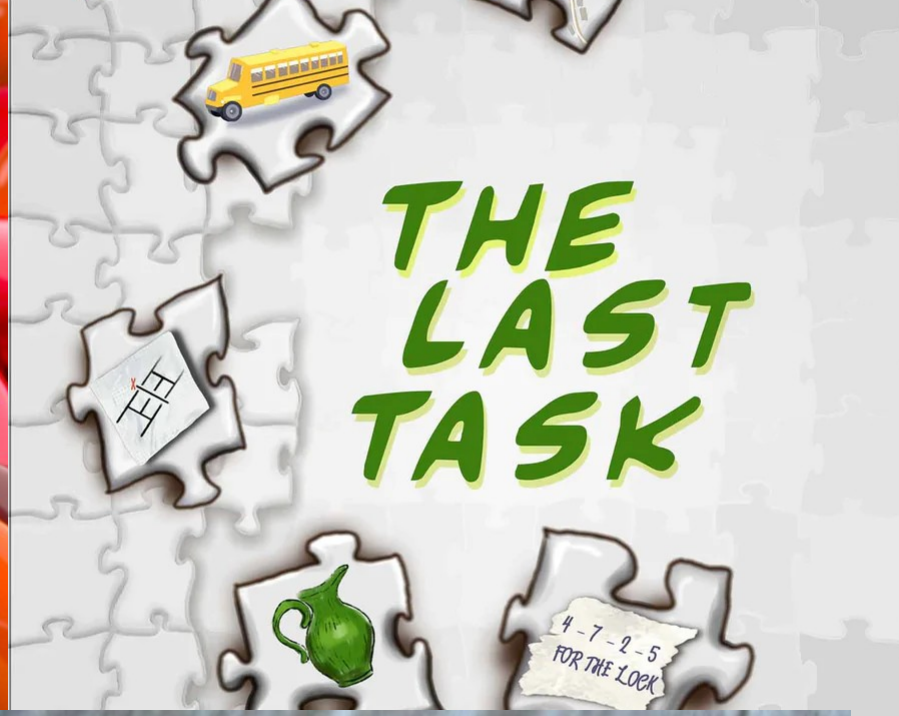
Check on Syllabus; & by appointment





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Option 1

Last Task

S3 Databucket

DELTA LAKE

Option A: Implement Delta Lakes for your Distributed File System to connect with your cloud computing operating system, including following features

1. **VPS:** your application should be running in your cloud VPS. **(1point)**
2. **Data:** from not a VPS, but using a data repository/storage bucket type (S3), from where the system running on your VPS, should access a remote data repository, geographically located in a different country. **(2points)**
3. **Connection:** use SSH RSA public and private keys to allow automated safe connection; **(1point)**
4. **Security:** don't allow server remote login as root using password. **(2points)**
5. **Documentation:** describe in a word or ppt file the steps/settings taken in your implementation; demo: have the system to read and add files to the remote data repository; **(1point)**
6. **Submission:** upload on D2L, , source code, documentation and a not zipped short video demo ~5 minutes, and ppt presentation. **(3points)**

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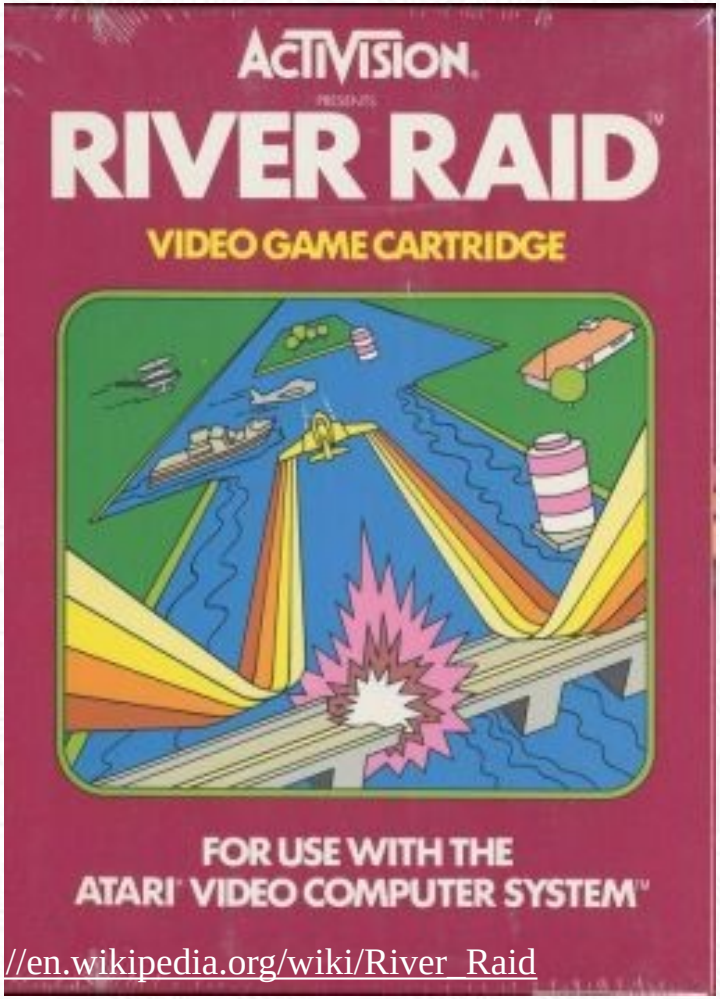
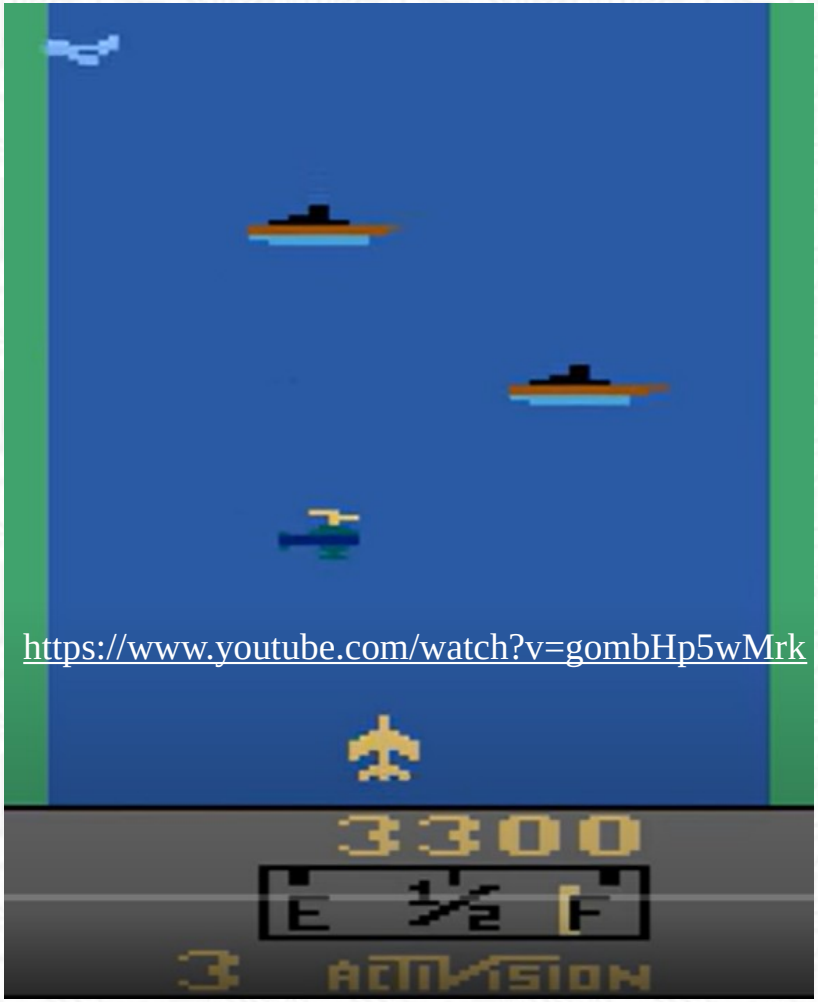
Last Task

River Raid Threads



//en.wikipedia.org/wiki/River_Raid



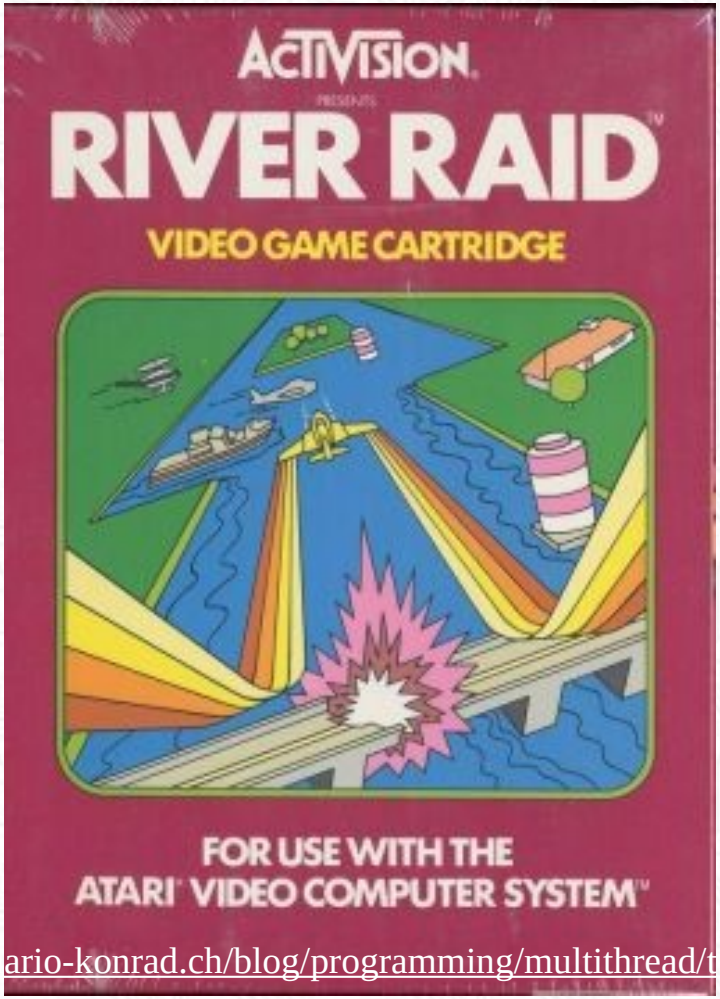
https://www.youtube.com/watch?v=gombHp5wMrk

Last TASK

TextStudio

River Raid Threads

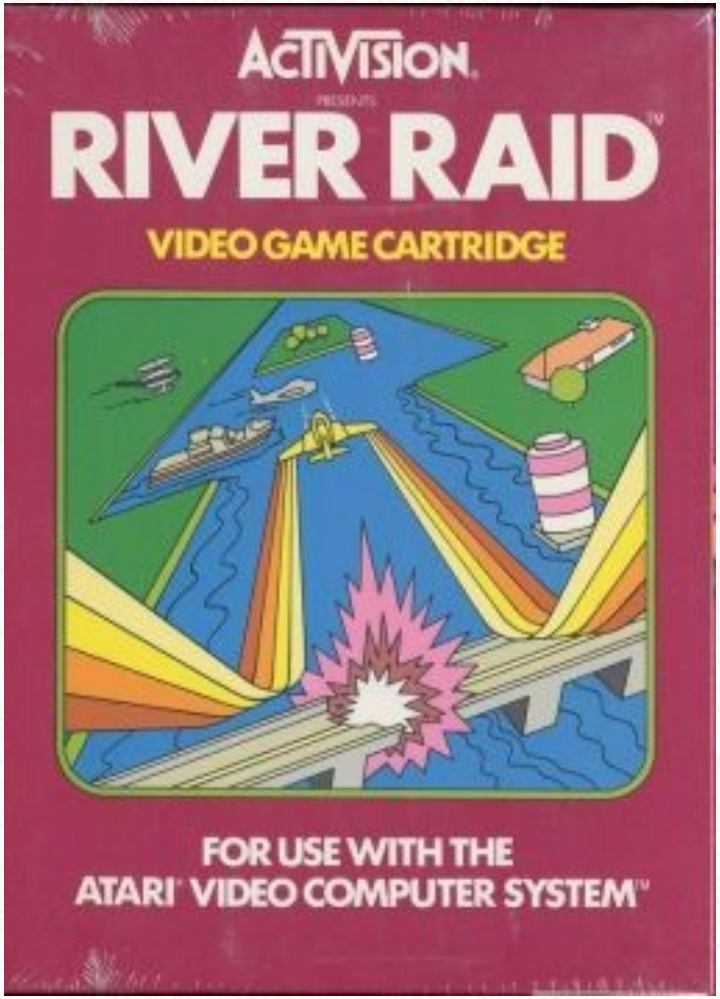
POSIX



ario-konrad.ch/blog/programming/multithread/t l-04.html

Last Task

River Raid Threads



Last TASK

River Raid Threads

Option B: Implement a program with threads with the features

1. **VPS:** your application should be running in your cloud VPS. **(1point)**
2. **Data:** from not a VPS, but using a client have the user running the pilot decisions. **(1point)**
3. **Connection:** use some SSH RSA public and private keys to allow automated safe connection and data transmission; the shared memory should be located on the VPS; The moves for the threads are automatically generated by computer (**H J B**), while the moves for the player **A** are controlled by the player. **(3points)**
4. **Security:** use some SSH RSA public and private keys to allow automated safe connection and data transmission. **(1point)**
5. **Documentation:** describe in a word or ppt file the steps/settings taken in your implementation; demo: have the system to read and write updates to the share memory located in the remote VPS; **(1point)**
6. **Submission:** upload on D2L, source code, documentation and a not zipped short video demo 5 to 10 minutes, and ppt presentation. **(3points)**